Kelly A. Loria

Postdoctoral Researcher, Desert Research Institute, Division of Hydrologic Sciences
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Education

2020 – 2025 **University of Nevada, Reno**
Ph.D., Department of Natural Resources and Environmental Science,
NSF Graduate Research Fellow

2014 – 2017 **University of Colorado, Boulder**
B.A., Ecology and Evolutionary Biology, College of Arts and Sciences,
Honors distinction in Research (Magna Cum Laude)

2013 – 2014 **Diablo Valley Community College**
Associate degree in environmental science

Academic Appointments

2025 – present **Postdoctoral Researcher, Desert Research Institute (DRI)**
Division of Hydrologic Sciences, Reno, NV

2017 – 2020 **Limnology Program Manager for Niwot Ridge Long Term Ecological (LTER) Program,** Boulder, CO

Selected Publications (*students mentored are underlined)*

Loria K, Lowman H, Krause J, Genzoli L, F, Harpold A, Chandra S, and Blaszczak J (2026). Mountain stream ecosystem metabolism and nitrogen cycling responses to hydroclimatic volatility. *Journal of Geophysical Research: Biogeosciences, 131*(10), e2025JG009229.

Lowman H, **Loria K**, Katona L, Krause J, Chandra S, MacIntyre S, Melack J, Naranjo R, and Blaszczak J (2026). Daily dissolved oxygen saturation patterns mediated by light and stream discharge in littoral regions of an oligotrophic mountain lake. *Aquatic Sciences, 88*(2), 66. https://doi.org/10.1007/s00027-026-01292-5

Loria K, Lowman H, Krause J, Katona L, Naranjo R, Scordo F, Harpold A, Chandra S, and Blaszczak J (2025). The influence of mountain streamflow on nearshore ecosystem metabolism in a large, oligotrophic lake across drought and wet years. *Limnology and Oceanography.* https://doi.org/10.1002/lno.70157

Scamardo J, Munger W, **Loria K**, Nauman B, Wang J, Leopold S, Heggli A, Huntly N, Baker M, and Meadow A (2025). Restoring Endangered Streams: The Science, Practice, and Law of Instream Process-Based Restoration Amid Climate Change. *Journal of Rangeland Ecology & Management.* https://doi.org/10.1016/j.rama.2024.08.032

Blaszczak J, **Loria K**, Krause J, Lowman H, Chandra S (2024). Making the smart watershed-to-lake connection: using high-frequency sensors and process-based aquatic ecosystem models to predict nearshore greening in Lake Tahoe. Water Quality Report to Nevada Division of State Lands.

Christianson K, **Loria K**, Blanken D, Caine N, and Johnson P (2021). On thin ice: Linking elevation and long-term losses of lake ice cover. *Limnology and Oceanography Letters.* https://doi.org/10.1002/lol2.10181

Loria K, McKnight D, Ragar D, and Johnson P (2020) The life aquatic in high relief: shifts in the physical and biological characteristics of alpine lakes along an elevation gradient in the

Rocky Mountains, USA. *Aquatic Sciences, 82*(1), 11. https://doi.org/10.1007/s00027-019-0684-6

Loria K, Christianson K, and Johnson P (2020). Phenology of alpine zooplankton populations and the importance of lake ice-out. *Journal of Plankton Research,* 42(6), 727-741. https://doi.org/10.1093/plankt/fbaa050

Presentations

(* indicates oral presentation)

*Loria K, Larsen W, Ryder J, Kruger B, Hausner M, Arienzo M. (September 2026). Variation in Seasonal Hydrologic Connectivity Regulates Post-Fire Solute Dynamics in a Mountain Catchment. MtnClim Conference.

Loria K, Larsen W, Ryder J, Kruger B, Hausner M, Arienzo M. (December 2025). First Flush, Future Impacts: Stream Water Quality Responses Across Burn Severity and Hydrologic Connectivity Following the Davis Fire. American Geophysical Union (AGU) Fall Meeting. Abstract B21H-1727.

Loria K, Lowman H, Krause J, Scordo F, Harpold A, Chandra S, Blaszczak J. (December 2025). Mountain Stream Ecosystem Metabolism and Nitrogen Cycling Responses to Hydroclimatic Volatility. AGU Fall Meeting. Abstract H51P-0553

*Loria K, Lowman H, Krause J, Katona L, Genzoli L, Harpold A, Chandra S, Blaszczak J. (May 2025). Spatiotemporal Variation in Mountain Stream Metabolism and Nitrogen Cycling Across Contrasting Flow Regimes. Society for Freshwater Science (SFS) Annual Meeting.

*Loria K, Lowman H, Krause J, Katona L, Naranjo R, Scordo F, Harpold A, Chandra S, Blaszczak J. (June 2024). Upstream Efficiency and Downstream Productivity: Linking Mountain Stream Processes with Nearshore Productivity in the Lake Tahoe Basin (California–Nevada, USA). SFS Annual Meeting.

*Loria K, Lowman H, Sudeep C, Blaszczak J. (June 2023). Mountain Streams' Role in Governing Physical Drivers of Nearshore Metabolism: A Case Study of Contrasting Flow Regimes in the Lake Tahoe Basin. Association for the Sciences of Limnology and Oceanography ((ASLO) Summer Meeting.

Loria K, Sudeep C, Blaszczak J. (May 2022). Seasonal Variation in Nitrate Uptake and Ecosystem Productivity Rates in Mountain Streams with Contrasting Flow Regimes in the Lake Tahoe Basin. Joint Aquatic Sciences Meeting (JASM). Poster presentation.

Loria K, Blaszczak J. (December 2021). Linkages Between Ecosystem Productivity and Nitrogen Uptake in Mountain Streams with Contrasting Flow Regimes in the Lake Tahoe Basin. AGU(AGU) Fall Meeting. Abstract H15I-1144. Poster presentation.

Loria K, Blaszczak J. (May 2021). Modeling Mountain Stream Metabolism: Methods for Linking Upland Rates of Biotic Nutrient Processing with Nearshore Productivity in the Tahoe Basin. SFS Virtual Annual Meeting. Poster presentation.

*Loria K, McKnight D, Ragar D, Johnson P. (2018). Comparative Survey of the Abiotic and Biotic Characteristics of Lakes Across an Elevation Gradient in the Rocky Mountains, USA. ASLO Summer Meeting.

*Loria K. (2017). The Effects of Elevation on the Plankton Communities of Alpine Lakes. Tenth Annual Guild of Rocky Mountain Ecologists and Evolutionary Biologists Meeting. Poster presentation.

Grants and Awards

- **2026** National Ecological Observatory Network (NEON) program Ambassador. Two years of support to augment engagement with aquatic scientist's use of NEON products to analyze environmental change.

- **2025** Desert Research Institute's Vice President's Office Award. *Leveraging machine learning to better predict how fire could impact stream water quality across the Western U.S.* (Lead PI, $20,000).
- **2024** NSF RII Track-2 FEC project supported student participation in trainings and conference proceedings through the NSF RII Track-2 FEC project Highly Predictive, Explanatory Models to Harness the Life Science Data Revolution, a collaborative effort among the University of Nevada, Reno, University of Wyoming, and University of Montana focused on developing computational and machine-learning approaches for high-dimensional life science data.
- **2024** MtnClim Student Scholarship to Tahoe City, CA for participation in the MtnClim Meeting ($350)
- **2024** Graduate Student Association Travel Grant to Philadelphia, PA for SFS Meeting ($500)
- **2023** Graduate Student Association Travel Grant to Palma, Spain for ASLO Meeting ($750)
- **2021** Graduate Student Association Travel Grant to New Orleans, LA for AGU Fall Meeting ($500)
- **2021** AGU Student Travel grant to New Orleans, LA for Fall AGU Meeting ($1000)
- **2021** Graduate Student Association Spring Research Award for Survey of Caldor Fire on stream water quality ($1,500).
- **2021-2022** The Southwest Climate Adaptation Science Center Natural Resources Workforce Development Fellowship ($8,000)
- **2021** National Science Foundation (NSF) Graduate Research Fellow (2020000930863) for project "Predicting stream ecosystem responses to climate change induced shifts in the hydrologic regimes of mountain watersheds"
- **2020** Honorable Mention in the National Science Foundation Graduate Research Fellowship Program competition.
- **2020** University of Nevada Reno Graduate Dean's Merit Scholar for students who show the potential to make substantive contributions to their discipline ($10,000).
- **2020** Whittell Forest Graduate Research Fellowship for University of Nevada Reno graduate research ($1,500).

Quantitative & Technical Expertise

- **Programming & reproducibility:** R, R Markdown/Quarto, python, Git/GitHub, reproducible data workflows
- **Statistical analysis: time**-series analysis, mixed-effects/hierarchical models, Bayesian modeling, trend analysis, concentration–discharge relationships, machine learning
- **Spatial analysis:** QGIS, spatial data processing, remote sensing data integration
- **Freshwater data:** high-frequency sensor data, streamflow, water temperature, dissolved oxygen, nutrients, DOC, suspended sediment, trace metals, ecosystem metabolism
- **Data products:** visualization, Shiny applications, automated data acquisition and processing, technical reporting

Teaching

Instructor of Record: GPHS 782: Watershed Science Seminar
University of Nevada, Reno; August-December 2026

- Designed and teach a graduate seminar integrating watershed hydrology, stream ecology, scientific literature, and reproducible R-based analysis of freshwater monitoring datasets with lab materials available on GitHub.

Teaching Assistantships at the University of Nevada Reno:

- NRES 295 Wildlands Hydrology **(**January-May 2021 and 2024)
- NRES 210 Environmental Pollution **(**January-May 2024),
- ENV 101 Introduction to Environmental Science (January-May 2021)
- NRES 422/622 Soil Physics (August-December 2020)
- NRES 322 Soil Science (August-December 2020)